

## Cartier divisors and line bundles

Let  $X$  be an integral scheme. For a Cartier divisor  $D$  locally given by  $f_i \in K(X)$ , we define  $\mathcal{O}(D)$  to be the  $\mathcal{O}_X$ -subsheaf of  $\mathcal{K}_X$  locally generated by  $f_i^{-1}$ . If  $X$  is regular in codimension 1 and regard  $D$  as a Weil divisor, it coincides with the sheaf

$$\mathcal{O}(D)(U) = \{f \in K(X), (f) + D \geq 0 \text{ on } U\}.$$

In this way Cartier divisors are in bijection with subsheaves of  $\mathcal{K}_X$  that are locally free of rank 1. Indeed, let  $L \subset \mathcal{K}_X$  be an invertible subsheaf. Let  $U_i$  be an open covering of  $X$  such that  $L|_{U_i}$  is free and generated by  $f_i \in K(U_i) = K(X)$ . Then  $D := (U_i, f_i)$  is a Cartier divisor and  $\mathcal{O}(-D) = L$ .

Since  $X$  is integral, every line bundle can be embedded to  $\mathcal{K}_X$ . The proof given in Hartshorne is quite abstract, which says a locally constant sheaf on an irreducible space is constant. This is achieved by looking at the stalk of generic point. In this way we see that an embedding of  $L$  to  $\mathcal{K}_X$  is nothing other than a nonzero rational section of  $L$ . To be explicit, let  $(s, U)$  be a nonzero rational section of  $L$ . Then for any rational section  $(t, V)$ , there is a unique rational function  $f$  such that  $fs = t$  on  $U \cap V$ . The map  $L \rightarrow \mathcal{K}_X$  is then given by  $t \mapsto t/s$ .

Therefore, a nonzero rational section of a line bundle gives a Cartier divisor. To be explicit, let  $(s, U)$  be a rational section of  $L$ . In trivializations  $U_i$ , let  $\alpha_i: L|_{U_i} \rightarrow \mathcal{O}_{U_i}$  be an isomorphism, then  $f_i = \alpha_i(s) \in K(U_i)$  gives a Cartier divisor. Indeed,  $\alpha_i$  and  $\alpha_j$  differs by an element in  $\mathcal{O}(U_i \cap U_j)^*$ . If  $s$  is a global section, then  $D$  is effective.

Warning: If  $L$  is an invertible subsheaf of  $\mathcal{K}_X$  and  $s$  is a section, then the divisor associated to  $s$  is not naively the divisor of the rational function.

## The canonical class

In this section  $k$  is a perfect field and  $X$  is a geometrically connected smooth curve over  $k$ . Let  $\omega_X = \Omega_{X/k}^1$  be the sheaf of differentials. It is locally free. For every prime divisor  $P$ , let  $e_P$  be a basis for the free  $\mathcal{O}_{X,P}$ -module  $\Omega_{X,P}^1$ . For example we can take  $e_P = dt_P$  where  $t_P$  is a uniformizer at  $P$ . Indeed  $df$  is a basis if and only if  $f$  is a uniformizer. The geometric intuition is clear, and an algebraic proof is given in Hartshorne II.8.7. Let  $\omega$  be a nonzero rational differential 1-form. Write  $\omega = fe_P$ . We define

$$v_P(\omega) = v_P(f).$$

This valuation is independent of the choice of  $e_P$ . Define the divisor  $(\omega)$  associated to  $\omega$  by

$$(\omega) = \sum_P v_P(\omega) P.$$

This is just the Weil divisor associated to the Cartier divisor given by the rational section  $\omega$  of the line bundle  $\omega_X$ . The divisor class is denoted by  $K_X$  and is called the canonical class.

## Duality

Let  $D$  be a divisor. Define

$$\Omega(D) = \{w \in \Omega_{K(X)/K}^1 - \{0\} : (w) \geq D\} \cup \{0\}$$

which is the global section of  $\omega_X \otimes \mathcal{O}_X(-D)$ .

Define a pairing between a rational form  $\omega$  and a repartition  $r = (r_P) \in R$  by

$$(\omega, r) = \sum_{P \in |X|} \text{Tr}_{k(P)/k} \text{Res}_P(r_P \omega)$$

The pairing has the following properties.

- $(\omega, r) = 0$  if  $r \in K(X)$ . This is the residue formula since  $f\omega$  is a rational form.
- $(\omega, r) = 0$  if  $\omega \in \Omega(D)$  and  $r \in R(D)$ .
- For every  $f \in K(X)$ ,  $(f\omega, r) = (\omega, fr)$ .

Denote  $\theta(\omega)$  the linear form on  $R$  given by

$$\theta(\omega)(r) = (\omega, r).$$

**Theorem 3.1** (Duality). *This pairing descends to a perfect pairing*

$$\Omega(-D) \otimes R / (K(X) + R(D)) \rightarrow k.$$

**Lemma 3.2.** *Let  $\omega$  be a rational differential 1-form and let  $D$  be a divisor on  $X$ . Suppose the linear form  $\theta(\omega)$  vanishes on  $R(D)$ . Then we have  $\omega \in \Omega(D)$ .*

*Proof.* Suppose  $\omega \notin \Omega(D)$ . Then there exists  $P \in |X|$  such that  $v_P(\omega) \leq v_P(D) - 1$ . Let  $r = (r_Q)$  be the repartition such that

$$r_P = \frac{a}{t_P^{v_P(\omega)+1}}$$

where  $a$  is an element in  $\mathcal{O}_{X,P}$  to be determined, and zero elsewhere. Then  $r \in R(D)$ . However,

$$\text{Res} \frac{\omega}{t_P^{v_P(\omega)+1}} \neq 0,$$

and since the trace pairing on a separable field extension is nondegenerate, we may choose  $a$  so that

$$\mathrm{Tr}_{k(P)/k}(a \operatorname{Res} \frac{\omega}{t^{v_P(\omega)+1}}) \neq 0,$$

and thus  $(\omega, r) \neq 0$ , a contradiction.  $\square$

*Proof of duality.* For every divisor  $D$ , denote by  $J(D)$  the dual space of  $R/(R(D) + K(X))$ . If  $D_1 \geq D_2$  then  $J(D_1) \subset J(D_2)$ . Let  $J$  be the directed union of all  $J(D)$ .

For  $f \in K(X)$  and  $\alpha \in J(D)$ , define  $f\alpha$  to be the linear form  $r \mapsto \alpha(fr)$  on  $R$ . Notice that  $f\alpha \in J(D + (f))$ . This makes  $J$  a vector space over  $K(X)$ .

Note that  $\theta$  maps  $\Omega(D)$  to  $J(D)$ , and therefore induces a map  $\Omega_{k(X)/k}^1 \rightarrow J$ . By the lemma above we know  $\theta^{-1}(J(D)) = \Omega(D)$ . It suffices to show  $\theta$  is an isomorphism. Clearly  $\theta$  is injective. By the following lemma,  $\theta$  is surjective.  $\square$

**Lemma 3.3.** *We have  $\dim_{K(X)} J \leq 1$ .*

*Proof.* Suppose  $\alpha, \alpha'$  are two elements in  $J$  that are linearly independent. Choose a divisor  $D$  such that  $\alpha, \alpha' \in J(D)$ . Let  $d = \deg D$ . Fix  $P \in |X|$  and let  $\Delta_n = nP$  for every integer  $n$ . If  $f, f' \in L(\Delta_n)$ , then we have  $f\alpha + f'\alpha' \in J(D - \Delta_n)$ . This gives a monomorphism

$$L(\Delta_n) \oplus L(\Delta_n) \rightarrow J(D - \Delta_n), \quad (f, f') \mapsto f\alpha + f'\alpha'.$$

So we have

$$\dim_k J(D - \Delta_n) \geq 2 \dim_k L(\Delta_n).$$

This is impossible, since by Riemann–Roch the left side grows like  $n$  and the right hand side grows like  $2n$ .  $\square$

*Remark 3.4.* The proof seems to be rather elementary. In fact, there are two key ingredients in the proof

1. residue formula
2. finiteness of cohomology.

For compact Riemann surface the residue formula is easily achieved, and the real difficulty lies in the finiteness of cohomology. For algebraic curves, the residue formula is hard.

*Remark 3.5* (Pairing in terms of line bundle). We describe the pairing of

$$H^1(X, L) \otimes H^0(X, L^\vee \otimes \omega) \rightarrow k$$

without writing  $L$  as  $L(D)$ .

Firstly,  $H^0(X, L^\vee \otimes \omega) = \text{Hom}(L, \omega)$ . By functoriality of cohomology,  $f \in \text{Hom}(L, \omega)$  gives a map  $H^1(X, L) \rightarrow H^1(X, \omega)$ .

By tensoring the exact sequence

$$0 \rightarrow \mathcal{O}_X \rightarrow \mathcal{K}_X \rightarrow \bigoplus_{P \in |X|} (K/\mathcal{O}_P) \rightarrow 0$$

with  $L$ , we get

$$0 \rightarrow L \rightarrow L \otimes \mathcal{K}_X \rightarrow \bigoplus_{P \in |X|} (L_\eta/L_P) \rightarrow 0$$

where  $\eta$  is the generic point.

By taking global sections, we get

$$L_\eta \rightarrow \bigoplus_{P \in |X|} L_\eta/L_P \rightarrow H^1(X, L) \rightarrow 0.$$

For  $f \in \text{Hom}(L, \omega)$ , by naturality we have

$$\begin{array}{ccccccc} L_\eta & \longrightarrow & \bigoplus_{P \in |X|} L_\eta/L_P & \longrightarrow & H^1(X, L) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ \Omega_\eta & \longrightarrow & \bigoplus_{P \in |X|} \Omega_\eta/\Omega_P & \longrightarrow & H^1(X, \Omega_X) & \longrightarrow & 0 \end{array}$$

Finally we define the trace map  $H^1(X, \Omega_X) \rightarrow k$  as the sum of residues. It is easily checked if  $L = L(D)$ , everything coincides with the previous construction of pairing.